2024 Syllabus

Foundations of Physics (PHYC10009)

Lectures:

Monday:	1.00 – 2.00	Lyle Theatre
Wednesday:	1.00 – 2.00	Lyle Theatre
Friday:	2.15 – 3.15	Lyle Theatre

Internet:

Canvas Learning Management System: http://www.lms.unimelb.edu.au/

Recommended Text:

Douglas C. Giancoli, *Physics: Principles with Applications Global Edition,*

Pearson, (7th edition), 2015

Subject Lecturers:

Professor Martin Sevior: martines@unimelb.edu.au (Week 1-6)
Professor Phillip Urquijo: phillip.urquijo@unimelb.edu.au (Week 7-12)

Timetable:

26th February	Week 1	Lectures commence
4th March	Week 2	Tutorials classes commence
11th March	Week 3	laboratory classes commence
29th – 7th April		Mid-semester break (no teaching)
8th April	Week 6	Lectures and classes recommence
24th May		Last day of semester
27th May - 31st May		SWOT VAC
3rd – 21st June		Exam period

General Information

Course Description

This subject is designed for students with a minimal background in Physics and aims to provide a foundation in physics understanding for a range of important physics principles and applications. Emphasis is placed on key concepts rather than detailed analysis.
Topics include:
Mechanics: describing and explaining translational motion (1-D and 2-D); laws of motion and forces; energy transfer; momentum and impulse; equilibrium in the context of human and animal movement and transport.
Waves: explore wave properties in the context of sound.
Electrostatics and Electricity: describe than explain the interactions between electrostatic charges; explore moving charges and electrical components such as batteries and resistors in simple electrical circuits.
Magnetism: describe the effects and origin of magnetic fields; explore the relationship between charges and magnetic fields in the context of induction and electrical motors.
Modern Physics: explore and describe the concepts behind our understanding of the subatomic structure of materials and basic quantum mechanics.
Radiation: explore and describe the different processes of radioactive decay.

Assessment

Assessment will be derived from the end-of-semester exam (60%), homework (online assignments, 15%), and laboratory work (labs, 25%).

Exam: A 3-hour exam will be set at the end of the semester. It will cover *all* of the material in semester 1. You will be provided with a formula sheet (made available before the end of the semester). You are expected to be available to sit an exam at any time during the University's exam period. The exam timetable will be finalized by central timetabling later in the semester.

Classes

Activity	**Weekly Occurrence**	**Hours - Total**
Lectures	3 per week	36
Lectures will introduce you to the content of the subject. It is expected that you supplement the lectures with reading the textbook and more widely, explore any resources, websites provided you and attempt any set problems.		
Problem-Based Classes	1 per week	12
Starting in Week 2, the Problem-Solving Classes will give you an opportunity to try questions with your fellow students under the guidance of a tutor. It is expected that you come to these classes having attempted the homework sheets already.		
Laboratory Sessions	1 per week over 6 weeks	15
You will have three laboratory sessions and three activities throughout the semester. At the end of each session you will submit a logbook entry detailing the work you did in the experiment. The experiments are compulsory and constitute a hurdle requirement for the successful pass of the subject. You must complete all of the experiments and obtain at least 50% grade overall.		

Outline

Note: This guide is an indication of the distribution of material. The lecturers may choose to cover some topics in more or less depth than is suggested here, and the order of presentation may vary to some extent.

Week	Lecture	Topics	Textbook reference / Questions
Week 1	1	Motion in one-dimension; velocity and speed; instantaneous velocity	2.1-2.3 **Q: Ch2:** 3, 6, 9, 10, 15*, 16*
	2	Graphs of motion; acceleration; motion with constant acceleration; free-fall motion; vectors and co-ordinate systems	2.4-2.7, 3.1 **Q: Ch2:** 18, 20, 25, 31, 35*, 38, 49*, 57
	3	vector algebra; two-dimensional kinematics; projectile motion	3.2-3.6 **Q: Ch3:** 3, 4, 9, 13, 18, 19, 26, 27, 32, 35*
Week 2	4	Introduction to dynamics; forces; Newton's laws of motion.	4.1-4.5 **Q: Ch4:** 1, 4, 6, 10, 16
	5	Specific forces; solving problems using Newton's laws; free-body diagrams	4.6, 4.7 **Q: Ch4:** 2, 11, 21, 26, 33*
	6	Friction	4.8, 4.9 **Q: Ch4:** 36, 37, 40, 49, 59*
Week 3	7	Circular motion; Rotational dynamics; static equilibrium; torque	5.1, 5.2, 8.4, 8.5, 9.1 **Q: Ch5:** 5, 8, 14 **Ch8:** 24, 26
	8	Gravity; orbits, weightlessness further applications of Newton's laws	5.6-5.8, 5.10 **Q: Ch5:** 27, 29, 40, 43, 49, 52*, 54
	9	Momentum and impulse; conservation of momentum	7.1-7.4 **Q: Ch7:** 1, 6, 10, 14*, 17, 22, 27
Week 4	10	Inelastic collisions. Energy; work and kinetic energy	7.6, 6.1 6.3 **Q: Ch7:** 35, 37 Ch6: 2, 5, 9, 16, 23
	11	Inelastic collisions. Energy; work and kinetic energy	6.4, 6.6-6.8 **Q: Ch6:** 28, 30, 32, 40, 45, 48*
	12	Oscillations, Hookes Law; Springs, Simple Harmonic motion	11.1-11.3 **Q: Ch11:** 3, 5, 11, 16, 56
Week 5	13	Waves, Longitudinal and Transverse waves	
	14	Sound waves; Loudness; Interference	12.1, 12.2, 12.9 **Q: Ch12:** 8, 11, 64, 65, 79,80
	15	Atoms, Light rays and waves' Ray Optics; Reflection; Mirrors	23.1-23.3 **Q: Ch23:** MC 2, 3; P: 10, 13, 17

Week 6	16	Refraction; Snells Law; Total internal reflection	23.4-23.6 **Q: Ch23:** 25, 27, 31, 32, 34
	17	Image formation; human eye	25.2 **Q: Ch25:** Q11 MC3, 6; P: 12, 14, 21

Lectures Week 7-12

Week 7	18-20	**Electrostatics:** electric charge and Coulomb's law; electric field, electric potential.	16.1, 16.5-16.8, 17.1-17.3, 17.5 **Questions:** **Ch16:** MC4, 9; Q1, 7, 14, 18, 21; 18, 22, 26 **Ch17:** 1, 3, 5, 7, 14, 17, 19, 2
Week 8	21-23	**Circuits:** electric current; resistance; power, electric circuits, resistors, basics of voltage/current. **Magnetism:** magnetic force, magnetism from current.	18.2-18.5, 20.1-20.2 **Questions:** **Ch17:** 34, 38, 40, 44, 48, 49 **Ch18:** 1, 2, 7, 11, 28, 31 **Ch19:** 1, 5, 10, 13, 21, 25 **Ch20:** MC 2, 3, 9
Week 9	24-26	**Magnetism:** magnetic force on moving charge; induction and Faraday's Law; transformers. **Spectra:** electromagnetic waves and colour.	20.3-20.4, 21.1-21.2, 21.7 22.1-22.3 **Questions:** **Ch 20:** 1, 9, 19, 21*, 42, 55 **Ch 21:** 2, 5, 11, 26, 29 **Ch 22:** 8,11,14,18 **Ch 27**: 11, 13, 19
Week 10	27-29	**Spectra:** photons, de Broglie wavelength, atomic structure, hydrogen atom; atomic spectra, lasers. **Radiation:** atomic nucleus, isotopes.	27.3-27.4, 27.8, 27.10-27.12 28.11, 30.1, 31.1-31.2 **Questions:** **Ch27:** 13, 27, 36, 40, 42, 47, 48, 50, 51, 53 **Ch30:** MC:4, P: 6, 10, 14, 15
Week 11	30-32	**Radiation:** nuclear decay, ionising radiation, half-life; interactions of radiation and matter, radiation safety; Radiation Imaging – X-rays, PET, SPECT, Ultrasound, MRI.	30.3-30.6, 30.8, 31.4-31.5 31.8, 31.9 **Questions:** **Ch30:** 23, 27, 28, 31, 38, 52 **Ch31:** MC: 9, 12, 13, 14, P: 40, 70, 71, Q: 19, 20, 21
Week 12	33-35	**Radiation:** Therapeutic uses of radiation, nuclear reactions, nuclear fission, nuclear fusion. **Power:** fossil fuel, solar, wind; large-scale energy storage. GAMSAT and exam style questions.	31.6, 31.1-31.3, 15.5, 15.7, 20.10, 21.5